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RESEARCH ARTICLE

A Structured Question–Answer Dataset for Large Language Models in Power System Dynamics and Operations

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ABSTRACT This paper introduces Power Grid Question and Answer (PowerGridQA), a curated question–answer dataset for large language models in power system dynamics and grid operations. The dataset combines content from authoritative sources, including scholarly articles, Textbooks and official North American Electrical Reliability Corporation (NERC) reliability publications. The resulting corpus of over 28,000+ entries divide into Power System Theory, NERC reports and technical documents, and a handful of reasoning type entries. The dataset is intended to support research on knowledge grounding, reasoning fidelity, and source-aware question answering for power-system applications. Each are technically verified Q/A pairs, emphasizing definitions, equations, and cause–effect relationships. PowerGridQA bridges the gap between traditional power system literature and modern AI models, supporting the development of factual, reliability-compliant language systems for real-world grid intelligence. A comprehensive analysis is done using a combination of traditional lexical metrics like Exact Match (EM), F – Score (F1), advanced semantic metrics like Bidirectional Encoder Representations from Transformers (BERTScore), and graph-based structural tests, including syntactic divergence analysis and thematic community detection. Furthermore, the structural analysis visually confirms the thematic coherence of the datasets through distinct topic clusters and illustrates the large syntactic gap between questions and answers in the reasoning subset.

INDEX TERMS Datasets, deep learning, electric power systems, information retrieval, large language models, machine learning, natural language processing, power system dynamics, power system operation, power system reliability, power system stability, retrieval-augmented generation, utility operations.

I. INTRODUCTION

Throughout recent years, the electric power grid is becoming increasingly complex as renewable energy integration, distributed energy resources, and evolving market dynamics introduce new operational challenges [1]. Grid operators must balance generation and load in real time parallelly while handling fluctuations, contingencies, and compliance with strict reliability standards [2]. So, even minor mistakes or knowledge gaps can contribute to instability or blackouts. There's a growing understanding that advanced AI and automation can play a major role in power system operations, stability assessment, and real-time diagnostics. These technologies can quickly process massive amounts of data and provide well-informed recommendations to operators. To help control room engineers analyze system behavior, test

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scenarios, and receive intelligent guidance, large language model (LLM)–based assistants are now being developed. These artificial intelligence (AI) systems are increasingly explored as decision-support tools, assisting operators by providing rapid access to procedural knowledge, documented best practices, and explanatory context for contingencies, faults, and stability-related concepts. Early demonstrations are promising like eGridGPT [3] which is an LLM designed to support power system operators with procedure analysis and scenario simulation in the control room [2], and research prototypes show that AI can extract decision-related information via natural language Q&A interfaces for dispatchers. Similarly, multi-agent systems that combine LLM reasoning with power flow solvers have been shown to automatically detect and mitigate grid violations (such as overloads or cyber-induced events) in real time [4]. These advances have illustrated the potential of AI to improve situational awareness, stability control, and response speed in modern power grids. However, realizing reliable AI-driven operations depends on the trustworthiness and domain competence of these systems. Importantly, such systems rely on tight integration with numerical simulators, grid models, or real-time telemetry, whereas the focus of this work is strictly on the textual knowledge layer required by language models operating in the power-systems domain. Power system operation is a safety-critical domain with specialized terminology, rigorous procedures, and physics-based constraints that general-purpose AI often fails to fully capture. One of the key limitations is that current high-performing LLMs have the lack of training on power grid-specific knowledge, so they tend to overlook or hallucinate crucial details of grid operations. This gap motivates the development of PowerGridQA which is a structured question–answer dataset focused on the power systems domain made to provide the high-quality, authoritative training and evaluation data needed for trustworthy Artificial intelligence in grid operations. The related work of similar type dataset for different operations and domains is given in Table 1.

Also, Experience in other technical fields has shown that domain-specific datasets are critical for training and benchmarking AI models that can reason reliably about specialized content. In the power systems arena, the availability of curated, expert-verified textual data (e.g. manuals, standards, and textbooks) is especially important because real operational data and incident logs are seldom publicly shared due to security and privacy concerns. Indeed, industry reports note that power sector data for AI is often proprietary or sensitive, forcing researchers to rely on synthetic scenarios or simulations. This lack of open data has been identified as a major barrier to developing robust AI for grid management. High-quality, domain-specific corpora can address this barrier by encoding the formal knowledge that grid engineers and operators accumulate through training and experience. For instance, PowerGridQA draws from authoritative textbooks like Handbook of Electrical Power System Dynamics [14] and North American Electric Reliability Corporation (NERC) documents [15]. These sources capture validated domain knowledge spanning from fundamental swing equation models to operating procedures and reliability standards, providing a rich basis for question–answer pairs. By structuring this knowledge into QA format, we create a resource that can be used to train and test AI models on operator-grade questions, including those involving grid equations, stability criteria, and emergency protocols. Notably, domain experts emphasize that formulas and technical rules are an essential part of power engineering knowledge that general Natural Language Processing (NLP) datasets rarely cover [12]. This level of detail is crucial for an intelligent system to generate responses that align with physical laws, and established engineering practice, even when operating purely at the language and knowledge-representation level. Ultimately, a high-fidelity QA dataset grounded in domain truth can serve as both a training substrate and a rigorous benchmark for AI models, helping to validate that they reason correctly and do not hazardously hallucinate in operational scenarios [1]. The intersection of AI and power systems has spurred several noteworthy efforts, but also highlights gaps that PowerGridQA aims to fill, Q&A fine-tuning is better than complete book fine tuning because it directly trains the model to answer questions, uses far fewer tokens, gives more control over answer structure, reduces hallucination, and avoids the memorization problems that come from full-book fine-tuning. Q&A data is cleaner, more efficient, easier to evaluate, and aligns closely with real user prompts, while full books mainly teach text continuation and contain a lot of irrelevant noise.

Despite this progress, there remains a conspicuous lack of publicly available, structured QA datasets in the power systems domain. Existing LLM-based solutions have largely relied on internally curated data or private utility information, and their evaluations use either general benchmarks or bespoke test sets that are not broadly accessible. To the best of our knowledge, no open dataset today provides a wide coverage of power engineering Q&A, especially with the level of detail (including grid operation scenarios, and NERC standards) required for operator-grade AI evaluation. In other scientific and technical domains, structured Q&A datasets have proven invaluable, from medical Q&A corpora for clinical decision support to formula-focused datasets for physics/math problem solving [10], [11]. But the power engineering community has not had an analogous resource. PowerGridQA addresses this gap by assembling a question–answer corpus from trusted power system literature and regulatory documents, ensuring that each QA instance is grounded in verified knowledge. We anticipate that this dataset will not only facilitate the training of reliable conversational agents for knowledge-grounded support in power-system contexts, but also enable rigorous benchmarking of their performance on realistic, high-stakes questions that grid operators and engineers encounter. The key novelty

TABLE 1. Related dataset table.

Dataset	Domain / Source	Size (questions / pairs)	QA format
PowerGridQA	Power-system dynamics, protection, stability, operations; textbooks + NERC Lessons Learned	≈28k pairs (23,368 theory, 4,696 NERC, 788 reasoning)	Short factoid, conceptual, and multi-step reasoning Q/A
SQuAD v1.1 [5]	Wikipedia reading comprehension	107,785 pairs	Extractive span-based QA
AQAD [6]	Arabic Wikipedia	17,911 questions	Extractive RC (span from paragraph)
SK-QuAD [7]	Slovak Wikipedia	91,165 questions	Extractive QA incl. unanswerable cases
BiQA [8]	Biomedical Q&A forums linked to PubMed	7,453 questions / 14,239 Q–article pairs	Document-retrieval QA
CyberMetric [9]	Cybersecurity standards & knowledge base	80 / 500 / 2,000 / 10,000 MCQs	Multiple-choice (4-option)
MedExQA [10]	Medical exam practice questions	965 MCQs	MCQ + explanation supervision
MedQA [11]	Medical licensing exams (USMLE/MCMLE)	61,097 questions	Multiple-choice QA
FormulaReasoning [12]	High-school physics word problems	5,324 QA pairs	Numerical reasoning with formulas
QASC [13]	Elementary science facts; crowdsourced questions + curated distractors	9,980 questions	8-way multiple-choice; each question linked to two supporting facts

of PowerGridQA lies in providing the first publicly available, source-traceable question–answer corpus specifically for the power systems domain that jointly covers textbook-level theory, real operational incident knowledge, and multi-step reasoning within a unified, structured QA format. To summaries the contributions of the dataset they include:

1. PowerGridQA, a publicly available, operator-relevant QA dataset for power system dynamics and NERC incident reports, formatted in JSONL for direct LLM training and evaluation.
2. A multi-layer characterization of the dataset (question types, topic communities, syntactic structure) showing that it stresses theory, operations, and multi-step reasoning rather than simple fact recall.
3. LLM and RAG baselines (EM/F1 and semantic metrics, FAISS-based retrieval on NERC events) and open-source all code and data to support reproducible benchmarking for power system-aware LLMs.

II. DATASET CURATION AND COMPOSITION

PowerGridQA is created to play a key role in building trustworthy AI for power system operations. It contributes to a well-organized and openly available dataset that helps researchers and engineers develop smarter tools for real-time diagnostics, grid stability analysis and operator decision-making. Fig 1 shows the construction work flow of the dataset

The Dataset consists of 28,000+ Q/A pairs which are split into around 23,368 question related to power system theory, resilience, stability, control operation and so on, these 23,368 question are taken from Textbooks and scholarly articles which are, An Introduction to Modelling of Power System Components [16], Electrical Power Systems [17], Analysis and Damping Control of Power System Low-frequency Oscillations [18], Communication, Control and Security Challenges for the Smart Grid [19], Power System Analysis [20], Power System Analysis [21], Power System Dynamic Stability [22], and Handbook of Electrical Power System Dynamics: Modeling, Stability, and Control [14]. These forms the fundamental and advanced knowledge base for the dataset which is the major bulk and back bone of the dataset. The sample pairs of this subset are shown in Figure 2. The next set for the dataset constitutes NERC Reliability questions and answer pair which are derived from NERC Lessons Learned Data Bank [15]. This contains around 4,696 Q/A pairs which are derived from each of the 108 currently published NERC lessons learned Quick reference guide handbook, it summarizes key problems, processes, and outcomes documented since 2015. The industry as a whole will be able to jointly identify problems, record what works, and exchange solutions thanks to these lessons learned. It's important because it clearly separates what actually happened from what was supposed to happen which will help organizations to avoid large-scale monitoring failures even if they stem from small issues like a firewall firmware bug or database capacity limits, or from major incidents such as a catastrophic battery energy storage system thermal runaway.

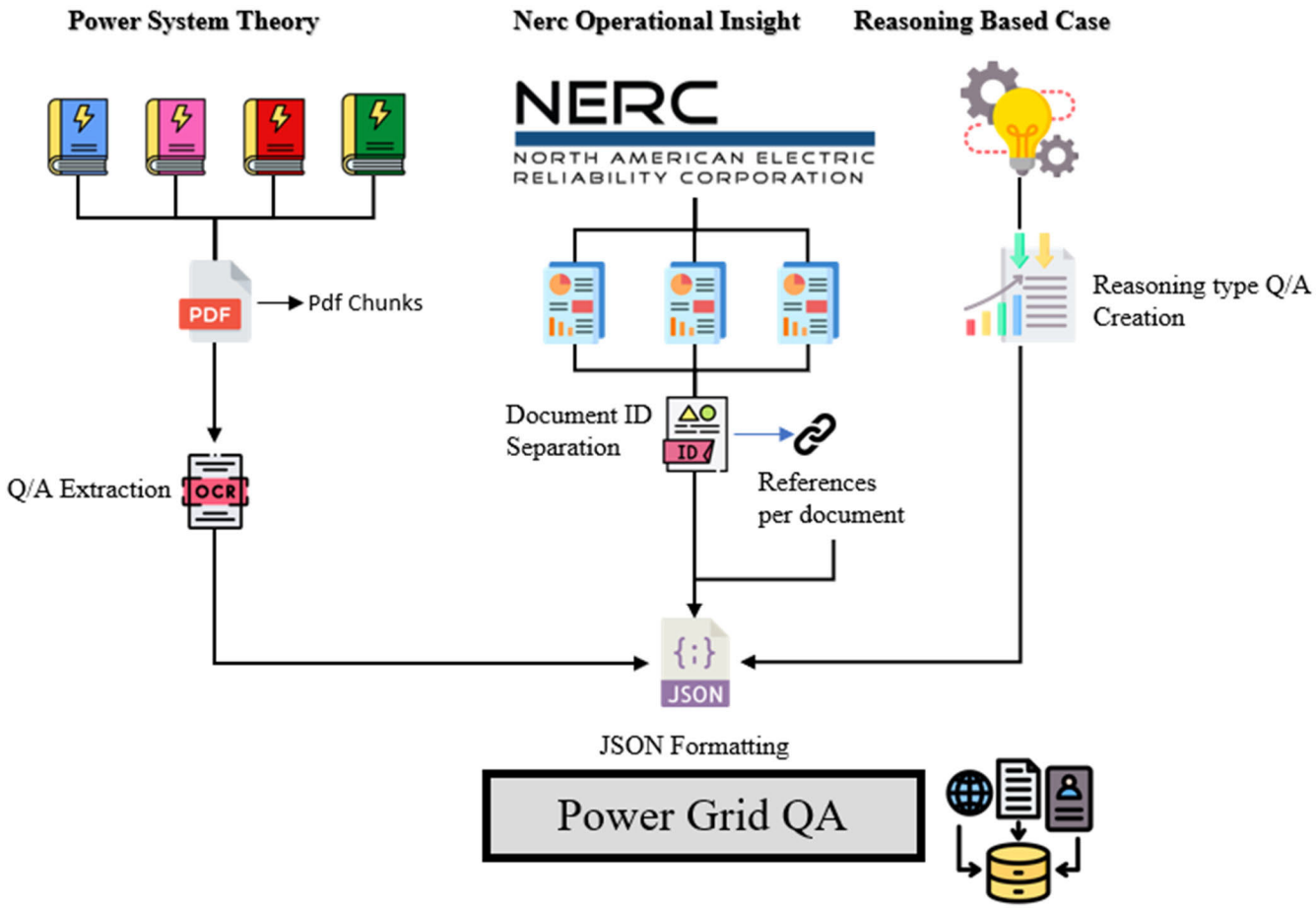

FIGURE 1. Dataset construction workflow for the Power Grid QA framework.

Context: Frequency stability in power systems depends on primary control action by turbine governors and on the overall system inertia. Governor droop determines how generation adjusts to arrest frequency deviations following a disturbance.
Question: What is the purpose of governor droop control in a power system?
Governor droop provides proportional frequency regulation by allowing generator output to change in response to frequency deviations. It ensures load sharing among units and prevents control hunting by introducing a steady-state frequency offset.
Question: How does reducing system inertia affect the Rate of Change of Frequency (RoCoF) after a generator trip?
A reduction in inertia doubles the RoCoF for the same power loss because less stored kinetic energy resists frequency deviation.

FIGURE 2. Extractive question–answer samples from the Power System Theory subset. Answers correspond to factual control and dynamic concepts taken from the dataset.

Context: In power system operations, relay protection schemes are designed to isolate faulty equipment to maintain grid stability. During a cold-weather event described in **NERC Lesson Learned LL20220801**, several SF_6 circuit breakers failed to operate because of low gas density alarms. As a result, operators implemented temporary logic blocking until gas density recovered.
What caused the SF_6 breakers to fail to operate?
Low gas density during cold weather.
Which operational measure was used to prevent mis-operations during low pressure?
Startup time delay logic blocking.
According to the Lesson Learned, what equipment characteristic was affected by temperature?
SF_6 gas density monitoring thresholds.

FIGURE 3. Example of question–answer pairs for a sample NERC Lesson Learned passage (LL20220801). Each answer is an extractive span from the passage, but domain-specific to power-system operational events.

This actively enhances the Bulk Electric System's (BES) dependability, resilience, and security by following this strategy rather than only responding to occurrences. The sample pairs of this subset are shown in Figure 3. Also, NERC Operational Insight subset is derived from publicly available NERC Lessons Learned reports and therefore reflects

operational practices, incident reporting conventions, and regulatory context specific to the North American bulk power system. This regional perspective is inherent to the source material and pertains specifically to the operational subset

The reasoning-based subset consists of a curated set of approximately 788 question–answer pairs designed to bridge the power system theory and NERC operational insight subsets. Unlike definition-based or single-fact queries, these questions are constructed to interconnect concepts across theoretical principles and real-world operational events, requiring answers to be derived through logical reasoning over multiple steps rather than retrieved directly from a single source.

This subset was purposefully designed to assess a model's capacity for true inference, such as causal reasoning and cross-context synthesis (e.g., applying protection theory to explain outcomes observed in NERC incident reports). Due to the relative rarity and importance of such complex inference scenarios in practice, the reasoning-based subset is intentionally smaller than the other subsets and is intended as a focused evaluation set rather than a large-scale training corpus.

At the same time, it is important to distinguish between the dataset's components: while the operational subset reflects NERC-specific practices, the power-system theory and reasoning subsets capture fundamental engineering principles (e.g., stability analysis, protection logic, control behavior) that are widely taught and applicable across global power systems.

The sample pairs of this subset are shown in Figure 4, our Q/A consist of all the three sets which will work interconnectedly to form an informed expert model.

The dataset collection was started with a deliberate focus on sources that combine both theoretical applications and real-world operational insights, the dataset was carefully constructed over several months carefully extracting from each of our data sources like the textbooks, scholarly articles and the NERC documents. Each book was split into 20-page pdfs focusing each on a particular topic on subject within the books, around an estimate of 50 question per pdf was extracted noting it cover the entire topics in that particular part of the book. These Q/A pairs where then verified and rechecked to ensure its importance and authenticity to ensure high-quality Q/A pairs. All extracted Q/A pairs were manually reviewed by domain experts with formal training in electrical and power systems engineering, including university lecturers, professors, and IEEE Senior Member/Fellow co-authors. Verification involved checking each answer against the original source for technical correctness and relevance. In cases of ambiguity or disagreement, the Q/A pair was revised through expert discussion to reach consensus. The selection of textbooks and source materials used for curation was also discussed and agreed upon by these domain experts prior to extraction the Q/A was strictly taken from the context of these reputed and authoritative source textbooks only, The question–answer pairs were initially generated through a structured, automated extraction

Context: Multiple NERC Lesson Learned reports describe cold-weather breaker failures involving different technologies.
Question: Based on NERC LL20221201 and LL202001101, both involving breaker failures in extreme cold, what is the key difference in the failure mechanisms?
LL20221201 was a mechanical/pneumatic failure—aged air-blast seals lost elasticity and leaked air.
LL202001101 was a thermodynamic failure—low SF_6 gas density triggered lockout.
Context: Power-system stability requires coordination between generator excitation systems and turbine governors to balance voltage and frequency control loops.
Question: Why can overly aggressive automatic voltage regulator (AVR) gains lead to system oscillations even when governors are properly tuned?
Because an excessively high AVR gain increases the effective electrical stiffness of the generator, reducing damping torque and amplifying electromechanical oscillations. Stable frequency control alone cannot suppress these voltage–angle interactions.

FIGURE 4. Reasoning-level examples from the Reasoning Subset. The first question synthesizes information across two NERC events while the second illustrates multi variable technical reasoning within generator control dynamics.

and transformation pipeline implemented in Python, which parsed the source material and formatted candidate Q/A pairs into JSONL Table 2. These candidate pairs were subsequently reviewed, refined, and corrected by domain experts to ensure technical correctness, relevance, and semantic abstraction rather than verbatim reproduction of the source text, with each question answer pair designed to be meaningful in itself. Table 3 shows the variation between each type of Q/A pairs.

Figure 5 shows the composition diagram of the dataset which gives the idea of what each of the subsets of the dataset contains, the formats have a simple but yet efficient structure with the addition of "Lesson ID" (Table 2) for NERC related pairs as its going to be easier to identify the exact case for an issue by identifying the corresponding ID.

Every answer was verified manually against the original text to eliminate speculation or hallucination. A strict review cycle followed: automated schema checks ensured consistent formatting, cosine-similarity filtering removed duplicates.

Table 3 provides examples on how each of the pairs are structured under each subset showing the difference in approach for each type of question pairs, the power system theory, NERC operational insights and the reasoning-based pairs are shown, noting the difference between each.

Examples of each type of question and the proceeding answer type is given so as to understand how each of the categories differentiate from each other giving an understanding of why each category complements each other, The power system theory for the general knowledge for the LLM,

TABLE 2. Q/A pair JSON format.

Category	Format
POWER SYSTEM THEORY, REASONING-BASED CASE	{"question": "", "answer": ""}
NERC OPERATIONAL INSIGHT	{"question": "In NERC Lesson Learned Lesson ID ", "answer": ""}

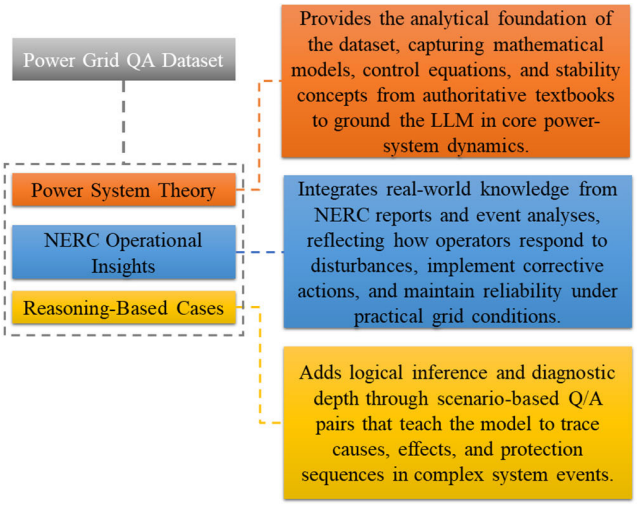

FIGURE 5. Composition of the PowerGridQA Dataset.

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III. DATASET ANALYSIS

This section aims to provide the complete data experiments, analyses or checks done in the dataset to support its technical quality. The over review of the tests and algorithms used is summarized in Figure 6. To provide a quantitative characterization of the dataset, an automated rule-based classification was performed using keywords [23] to determine the distribution of the question types within each.

The results which are visualized in Figure 7 reveals the distinct and specialized "fingerprints" for each dataset, validating their intended purposes. The results are shown in Table 4 as well for better understanding.

A more precise distribution for every sets of data is provided by this categorization, the Reasoning-based Case dataset is characterized by a large percentage of complicated Causal (22.8%) and Factual (50.4%) questions., the NERC Operational Insight dataset, on the other hand, continues to present a problem for rule-based approaches. With its sizable "Other" category (69.9%) showing its usage of varied, domain-specific vocabulary which can also be the IDs present in each of the pairs.

The Power System Theory dataset includes a nearly equal number of factual and causal questions which represents as a fair general-knowledge benchmark. This investigation confirms that every dataset is tailored for assessing various aspects of a model's performance.

To examine the thematic hierarchy within the dataset a term co-occurrence graph (Nodes on the graph are technical terms of interest (lemmatized nouns and proper nouns) was created and an edge exists between them whenever they appear in at least five question-answer pairs. Such a visualization helped us determine the central conceptual areas that the dataset addresses [24]. The obtained graph shows the presence of a number of homogeneous and consistent topical clusters, which confirms the selection of the dataset to the important aspects of the reliability of the power system.

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To perform part-of-speech (POS) tagging, spaCy's (an open-source software library for advanced natural language processing) [25] pre-trained model (en_core_web_sm) was used. It helps in filtering the words, keeping only nouns (NOUN) and proper nouns (PROPN) which will help the graph focus on the core subjects and entities. Then It removes common stop words (like "the", "is", "a") and non-alphabetic tokens. Afterward It uses lemmatization (token.lemma_) to get the base form of words (e.g., "systems" becomes "system"), grouping related terms.so, It returns a list of unique key terms for each document. This was used to create a term co-occurrence graph for the dataset subsets.

One of the most pronounced clusters focused on System Protection, and there is a strong correlation between such

TABLE 3. Representative Q/A pairs from the dataset.

CATEGORY	QUESTION	ANSWER
Power System Theory	What is the role of the excitation system in damping torque analysis?	The excitation system regulates generator voltage and reactive power, influencing damping torque and contributing to overall system stability.
NERC Operational Insight	In NERC Lesson Learned LL20231101, how did operators maintain situational awareness during the outage?	They used telemetry from neighboring transmission operators (TOs) and manual verbal updates from field personnel to estimate system state.
Reasoning-Based Case	A 345 kV line protection scheme fails to operate for a B-phase-to-ground fault caused by a 146 kA lightning strike. Both the System 1 and System 2 relays at one terminal are found to have rebooted at the exact moment of the fault. The relays are powered from the same station battery, but through separate fuses and wiring. Based on NERC LL20210204, what is the most probable cause of the simultaneous relay failure?	The most probable cause is a high-magnitude transient voltage induced onto the DC power-supply wiring by the lightning strike. Even though the DC supplies were separate, their wiring likely ran in parallel, allowing the transient to couple into both relay power supplies simultaneously, causing them to shut down and reboot.

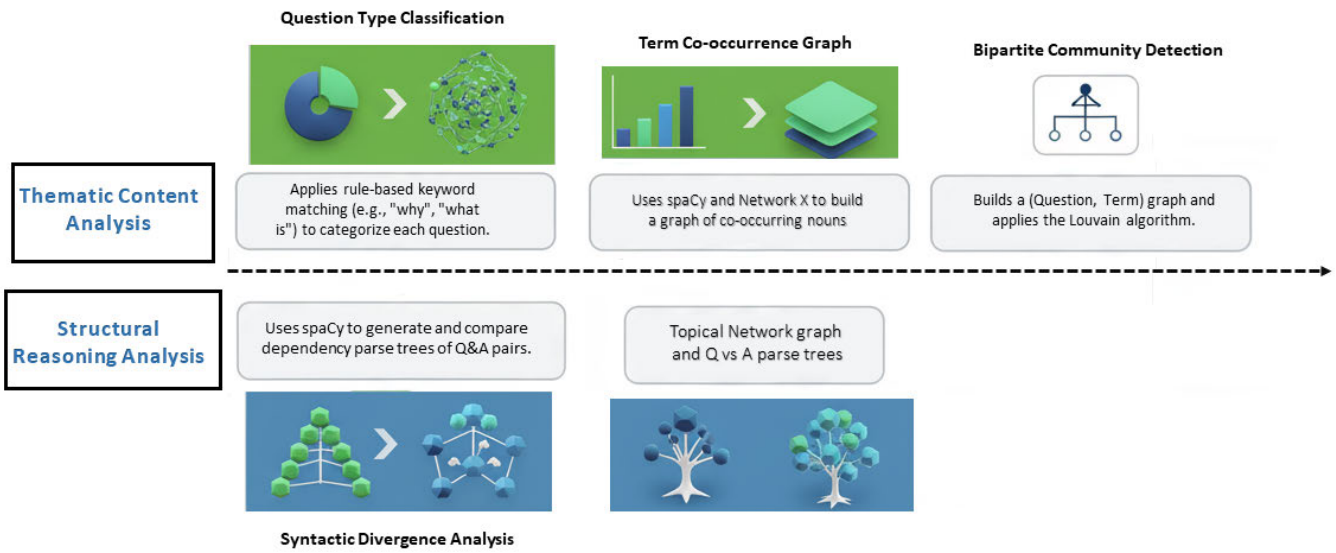

FIGURE 6. Overview of the analytical framework, showing the sequential steps for Thematic Content Analysis and the process for Structural Reasoning Analysis.

words as relay, setting, protection, and fault. The other significant group is connected to Generation and Grid Stability, where the concept of generator, voltage, frequency and plant is interrelated. A third cluster emphasizes Control Center Operations, which links such terms as operator, control, alarm and display.

Such clearly defined and domain-specific clusters evidences that the dataset is thematic and organized around the most important concepts in the Power system concepts and NERC event analysis. This implies that this inherent structure is a good and challenging benchmark to be used to train and evaluate models in terms of their capacity towards understanding complex domain specific relationships.

The "Power System Theory" graph Figure 8, shows a classic "core-and-spoke" structure it can be observed that the middle is a thick interlinked layer of core words with such words as system, power, generator, and line.

This thick center is the main ideas of power systems engineering. All of this is interrelated in that all subjects (such as voltage, stability, or control) are constructed using the same basic building blocks in theory. This chart is an ideal representation of a generalized academic body of knowledge. It becomes like a textbook, and in the end, chapters come to the same central principles.

This second graph Figure 9, representing the "NERC Operational Insight" dataset, is dramatically different.

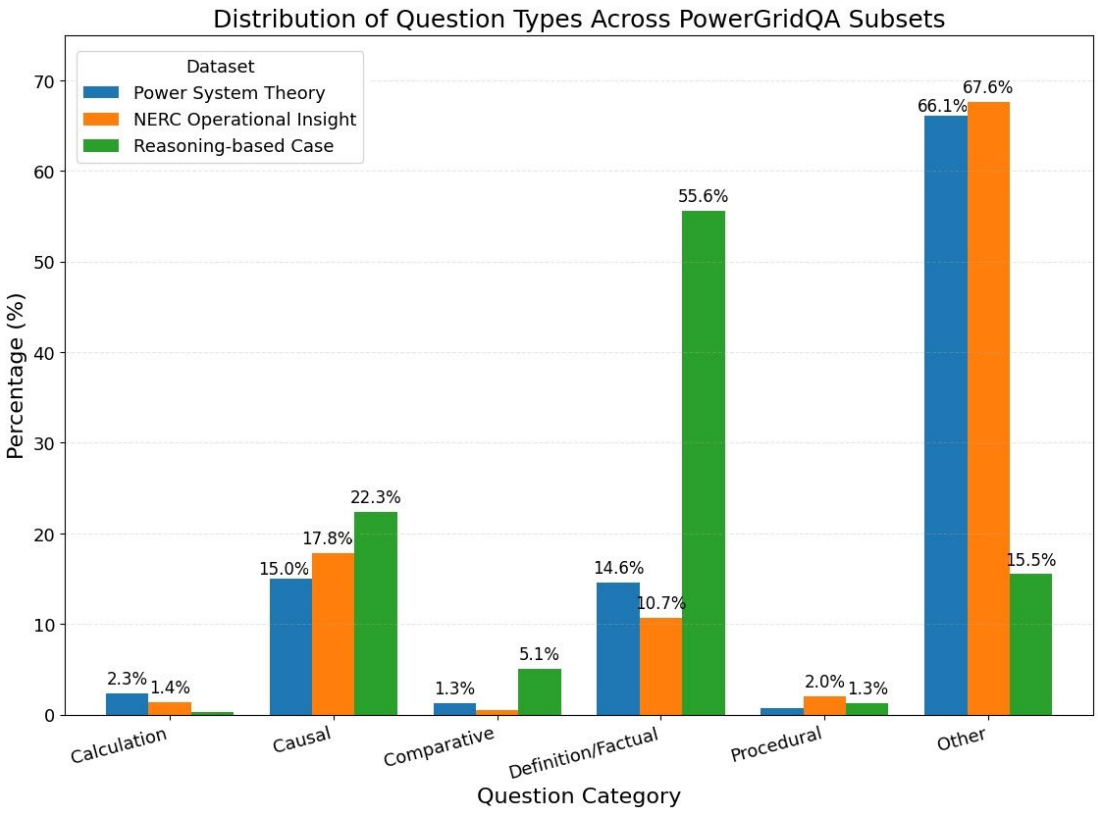

FIGURE 7. Distribution of Question Types Across Datasets.

TABLE 4. Distribution of question types across datasets tabular.

Question Category	Power System Theory (%)	NERC Operational Insight (%)	Reasoning-based Case (%)
Causal Reasoning	23.72	16.60	22.78
Definition/ Factual	23.52	9.82	50.38
Procedural	4.93	1.92	2.28
Comparative	0.61	0.32	6.84
Calculation/ Quantitative	0.99	1.40	0.76
Other	46.2	69.93	16.96

It doesn't have one giant, dense core. Instead, it appears as a galaxy comprising of multiple star clump.

The clusters below show the particular, practical theme or event type that can be inferred into:

- An "Event & Protection" cluster linking terms like relay, fault, and trip.
- A "Generation & Stability" cluster with generator, voltage, frequency, and plant.
- An "Operations & Control" cluster connecting operator, alarm, display, and control.

To gain a high-level map of the content in the Power System Theory dataset using a thematic analysis with bipartite community detection was used. This data strategy connects all the 27,080 questions with the 2,624 distinctive key words so that an algorithm can automatically cluster the questions into thematic groups using their overlapping words.

The dataset was utilized to identify the main ideas and connections between them. First, key terms were extracted from each question consisting of mainly nouns and proper nouns. Then, after cleaning, lemmatizing, and removing stop words. Afterward, a bipartite graph was built where one set of nodes is used to represent the questions and the other represents the unique terms. Each question is linked only to the terms it contains. Using NetworkX's greedy modularity algorithm, communities in this graph which are essentially clusters of questions that share similar key terms were detected [26]. Finally, these clusters were analyzed, sorted them by size,

and for the five largest ones, simple topic labels were created based on their most frequent terms and included sample questions to represent each cluster.

The analysis has managed to find unique clusters, which suggests an abundance and a diverse topical environment with general, more basic themes and smaller specialized niches. Table 5 summarizes the five clusters, the largest ones, and their central part of the dataset. These groups are well defined on major power systems engineering subjects.

To validate the coherence of these algorithmically-defined clusters, a sample from two of the largest and most distinct communities were visualized: Cluster 3 (Fault & Protection) and Cluster 4 (Load & Control).

In figure 10 (Cluster 3) the cluster addresses topics relating to electrical load management and controlling parameters associated with the system, likely only in their distribution networks. The questions in this group are made to address topics such as the voltage control methods, load shedding techniques, frequency control, and the challenges associated with managing power flow in distributions systems.

The cluster featured in figure 11 (Cluster 4) is geared towards the evaluation of electrical faults and the associated systems put in place to reduce their impact, the questions that can be found in this grouping will address issues such as relay settings, transformer protection schemes, fault current calculations, and line impedance relative to protection systems.

1) IMPLICATIONS FOR DATASET QUALITY AND COVERAGE

The thematic clustering and co-occurrence structures provide diagnostic evidence about PowerGridQA beyond visualization. First, the emergence of domain-consistent clusters (e.g., protection/fault, voltage/stability, and control/operations) serves as a coherence check, confirming that extracted QA pairs are organized around recognized power-system subdomains rather than forming a noisy or random term space. Second, the cluster-size distribution (Table 5) reveals coverage skew: large clusters indicate well-represented topic families (e.g., stability/voltage/protection), while smaller clusters highlight comparatively underrepresented niches (e.g., dispatch/PSS-related questions), which we explicitly acknowledge as coverage gaps and a priority for future dataset expansion. Finally, the contrast between the core-dense structure in the theory subset (Figure 8) and the multi-cluster "event-driven" structure in the NERC subset (Figure 9) supports the intended design distinction: the theory subset concentrates shared fundamentals, whereas the NERC subset is distributed across heterogeneous operational scenarios.

A syntactic divergence analysis was used to measure the reasoning challenge presented by this dataset. A pre-trained spaCy language model (en_core_web_sm) was utilized to analyze grammatical structures in the dataset. Firstly, a specific question and answer pair was selected from the dataset. then the spaCy pipeline was used to processes both texts which performed the tokenization, part-of-speech tagging, and dependency parsing on the pair to identify grammatical relationships such as subject–verb and modifier–head connections, then by using spaCy's displacy.render function with style ="dep", the dependency trees for both the question and answer are visualized, where arrows represent syntactic dependencies and labels denote grammatical roles like nsubj (nominal subject) or dobj (direct object). This process is done by creating and drawing the comparison of the dependency parsing trees of a question and answer to visualize the difference between its grammatical complexity Figure 11 and Figure 12 depicts the graphs between the dependency parsing of both the questions and answer.

- The Question (Figure 12):
  - Form: This is direct, simple and flat grammatical structure.
  - ROOT: The main verb is "is.
  - Main Clause: The main clause of the sentence is What is the risk. The remaining part of the sentence (of operating...) merely qualifies risk.

TABLE 5. Top 5 thematic clusters identified in power system theory dataset.

Cluster ID	Size	Top Defining Terms	Sample Question
Cluster 1	4,666	`system`, `stability`, `power`, `generator`, `contingency`	"How does the Z bus method help in evaluating transmission network robustness...?"
Cluster 2	3,848	`voltage`, `system`, `power`, `line`, `loss`	"Why is the transmission-loss equation important in economic operation?"
Cluster 3	3,667	`fault`, `protection`, `system`, `power`, `transformer`	"How are transformer equivalent circuit values obtained?"
Cluster 4	3,630	`load`, `control`, `system`, `power`, `distribution`	"What challenges exist in implementing voltage control in distribution systems?"
Cluster 5	305	`dispatch`, `challenge`, `system`, `power`	"What are the challenges in designing PSS stabilizers for multi-machine systems?"

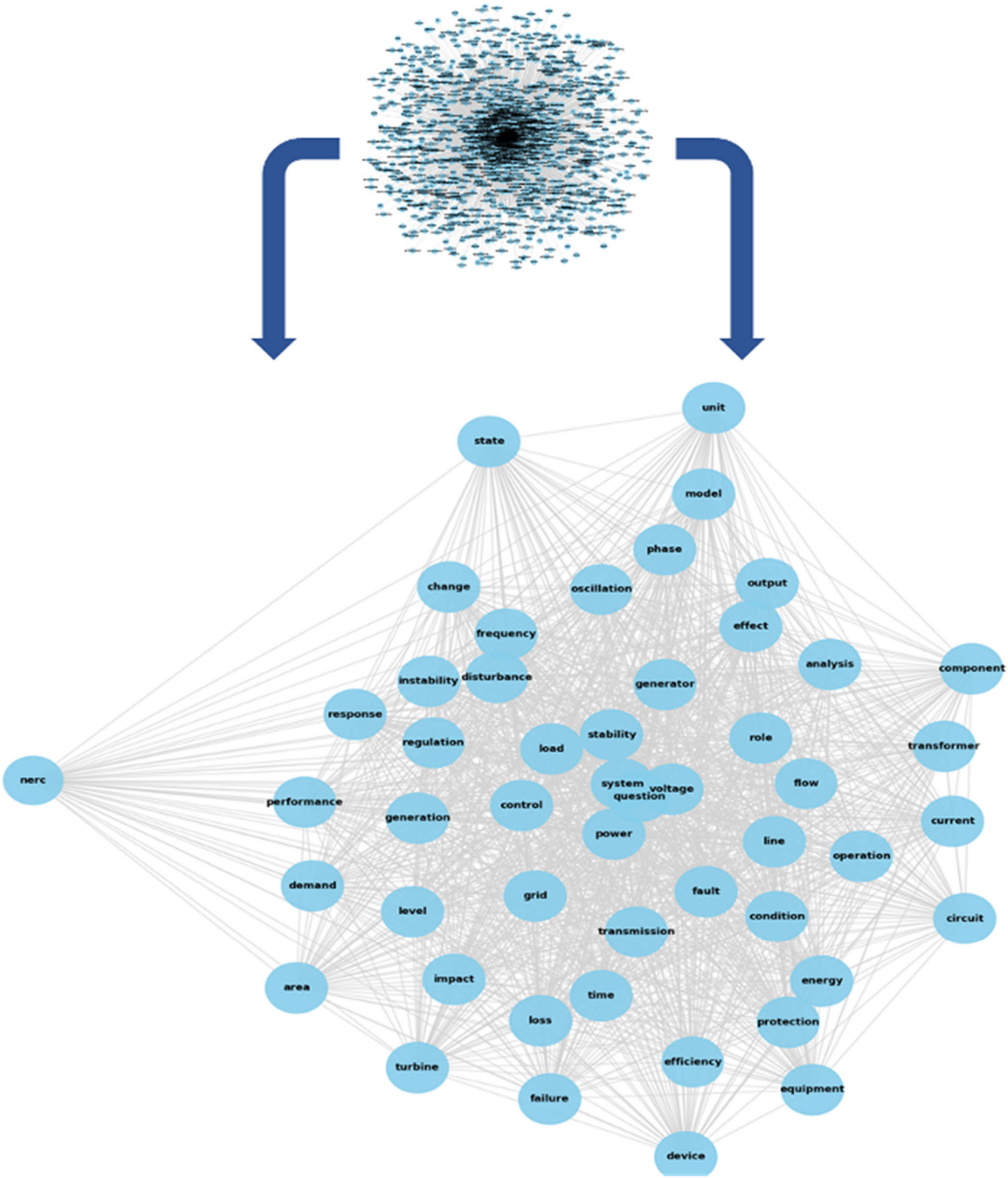

FIGURE 8. Power system theory term co-occurrence graph.

- Interpretation: It is simple What is X? question. It is grammatically simple and has one main clause.

- The Answer (Figure 13):
  - Form: It is the multi-layered, complex, and highly-nested grammatical form.
  - ROOT: The main verb is "Lead".

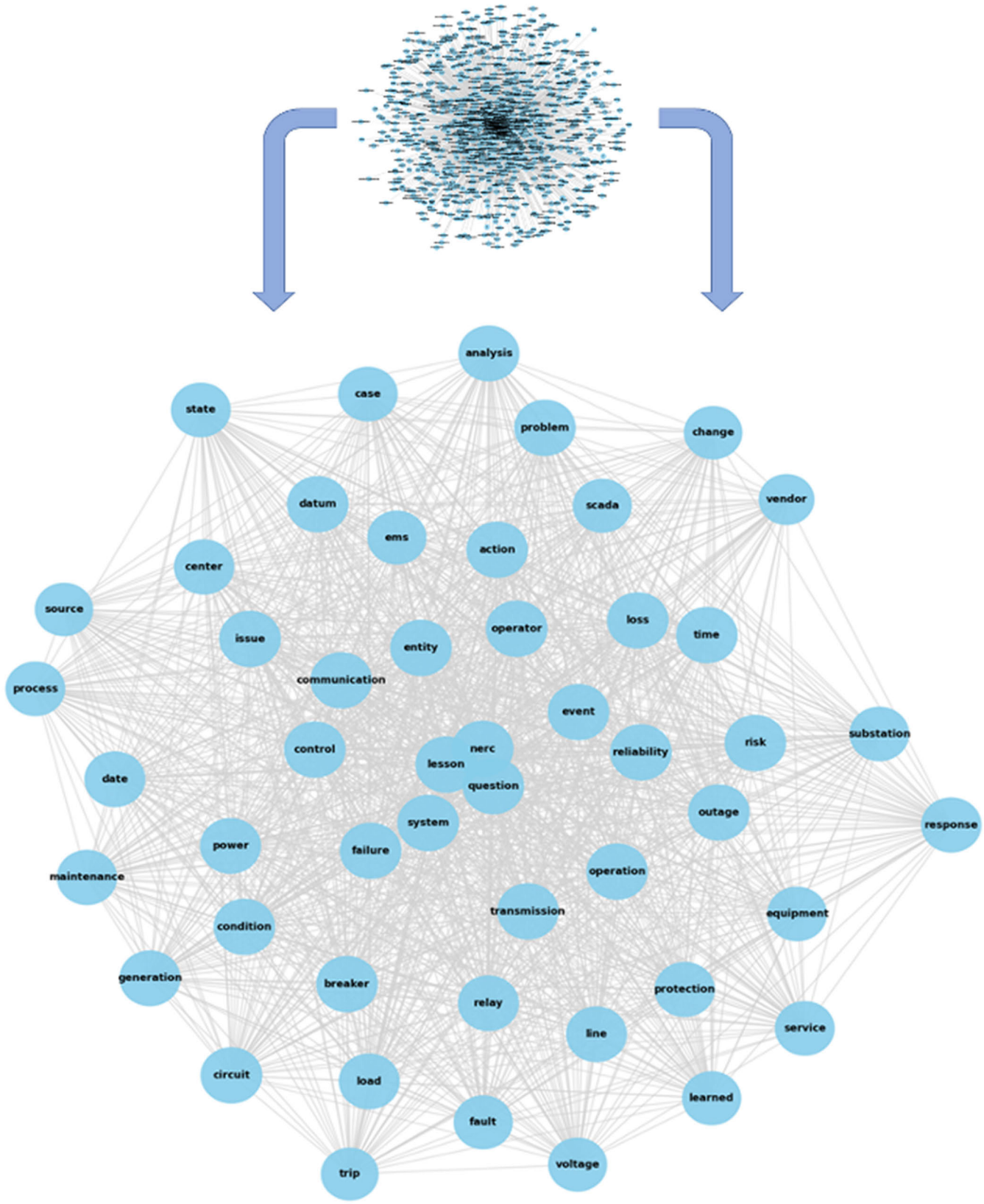

FIGURE 9. NERC Operational Insight term co-occurrence graph.

- Main Clause: The basic statement is that the operation of a generator can cause instability of voltages.
- Interpretation: It is a complex one due to the clauses within a clause.

Notice the word risking. It is linked to an advcl (adverbial clause modifier) label that adds a new clause entirely that explains the result of the instability of the voltage.

Then take note of the word mitigated. It is also linked with an advcl label, to form yet another subordinate clause (if not properly mitigated...) which specifies the circumstance in which the blackout takes place.

Through the analysis, there is a large structural difference between the query and the desired output. The question is "What is the primary risk of operating a generator with a compromised exciter?" has an uncomplicated, straightforward grammatical composition and a foreign primary clause. Contrastingly, the syntax of the answer is highly complex and buried. It has several dependent adverbial clauses (advcl) which present a series of consequential and conditionals logic which are of the original fault to possible voltage instability, and, finally, a regional blackout. This structural gap indicates that many questions are syntactically simple prompts, while correct answers often require multi-clause explanatory outputs (causal links and conditional qualifiers). This supports interpreting the dataset as explanation-heavy rather than factoid-only, and helps explain why strict lexical metrics (EM/F1) can remain low even when semantic similarity is high. We emphasize that syntactic complexity is used here as a proxy for linguistic/explanatory complexity, not as evidence of physical correctness or operational validity.

IV. EXPERIMENTS

The baseline performance measure was based on a standard evaluation script like the ones used in Stanford Question Answering Dataset (SQuAD) [5], [27] to compute the old-fashioned metrics, including Exact Match (EM) and F1 Score, and the semantic metric, BERTScore [28]. In order to view objectively this, an assessment was performed over the entire scope of each set of data. These results are shown in Table 5.

What struck us immediately was the stark difference between the lexical scores (EM and F1) and the semantic scores (BERTScore).

- The exact words of the reference answer (very low EM and F1) were not frequently used in the model in the case of Power System Theory, but the high BERTScore (82.74%) indicates that the model often captured the correct semantic intent despite failing to match reference phrasing, highlighting the prevalence of paraphrasing and concept-level reasoning in the dataset. This made us aware of the fact that this big test was not merely the attempt at trying to spot some keywords but that it entailed slightly larger amounts of paraphrasing and conceptual knowledge.
- The information on NERC Operational Insight provided a profile of what can be deemed as typical of technical information extraction. The nearly zero EM with moderate F1 (34.54) is an indication that the model was not bad at determining the right entities but bad at determining the ideal boundaries- a shortcoming arising when determining some NERC terminologies. It scored very high in BERTScore (84.36) that indicates that there was semantic accuracy in responses.
- The Reasoning-based Case dataset proved to be the toughest lexically, scoring 0% EM and only 20.29% F1. This wasn't surprising, given its design. However, there is a curious high BERTScore (85.40) that suggested that even though not all the logical chain was constructed, the model was able to identify the concept behind most of them correctly.

Closed-book models exhibit limited factual correctness despite increasing model size shown in Table 6, factual correctness is a measure of how accurate the responses are form the models. In contrast, retrieval over PowerGridQA enables a small 0.5B-parameter model to achieve substantially higher factual accuracy, highlighting the value of domain-specific knowledge grounding.

Without retrieval, instruction-tuned models frequently refuse to answer or fail to produce numerically correct responses on NERC operational questions as shown in Table 7. Numerical fact accuracy is hwne the model is able to look up the exact id and the corresponding informations from the NERC lessons. When augmented with reference-locked retrieval over PowerGridQA, the same base model produces grounded, numerically correct answers with substantially higher semantic alignment.

As the NERC Operational Insight data has a number of questions that contain document identifiers (e.g., NERC Lesson Learned IDs), it was anticipated to be particularly well matched with Retrieval-Augmented Generation (RAG) [29] systems. To experimentally test it, a semantic search experiment of retrieval accuracy was conducted in a standard semantic search paradigm.

We focus on retrieval-augmented evaluation rather than fine-tuning, as retrieval preserves source traceability and avoids memorization of domain-specific material, which is an important consideration in safety-critical power-system applications. Our objective is to assess how access to PowerGridQA affects model behavior and factual grounding, rather than to optimize a specific training pipeline.

A Facebook AI Similarity Search (FAISS) vector index [30] of the document embeddings (based on the question-answer text of each entry), was created using the all-MiniLM-L6-v2 [31] sentence transformer model into the document embeddings. this index was run against the original text of the question as used against each of the 4396 records that had a valid NERC ID and evaluated whether it would be feasible to get the correct source document back. Table 6 and 7 shows the evaluations done and compared with each subset for power system theory and NERC, results show substantial improvements than other baseline models.

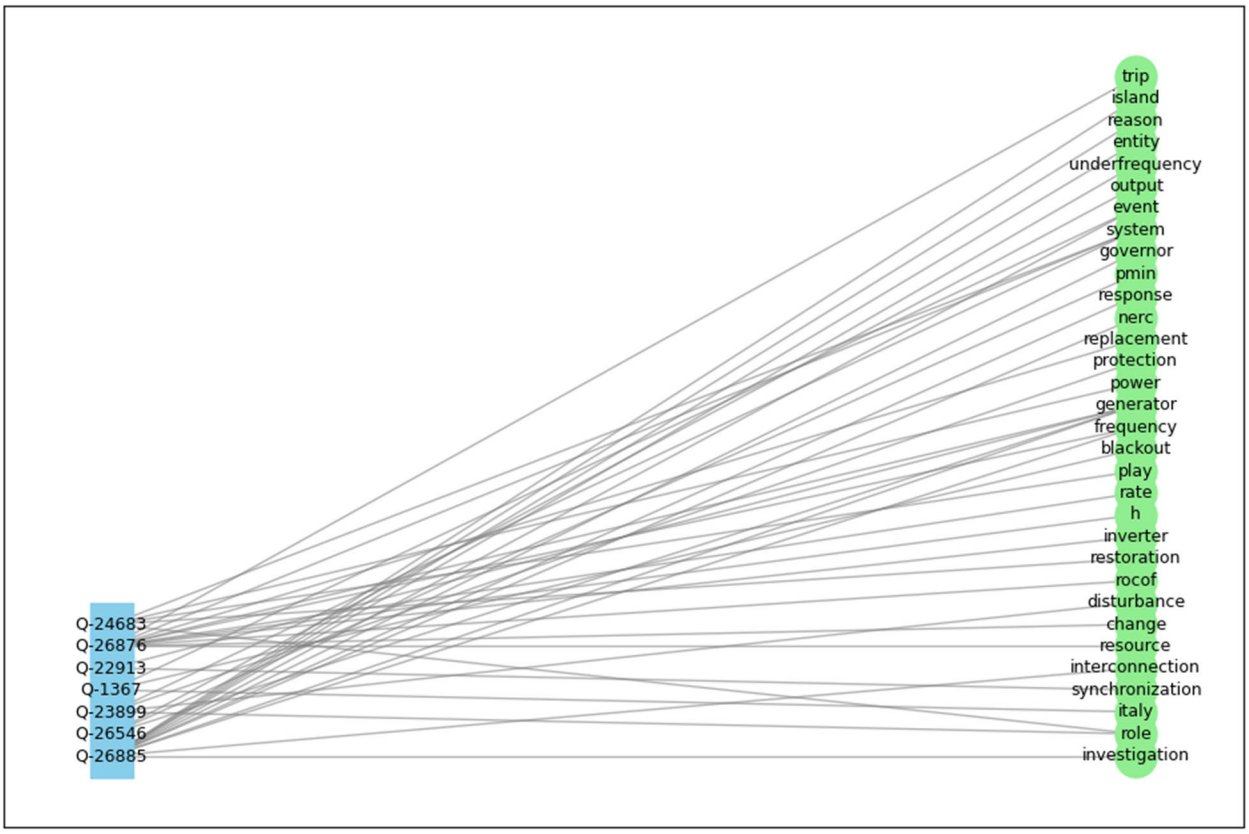

FIGURE 10. Cluster 3 represents System Protection and Fault Analysis.

TABLE 6. Closed-book benchmark comparison.

Model	Params	Avg. Similarity	Factual Correctness
Qwen2.5-0.5B-Instruct	0.5B	0.609	46%
LLaMA-3-3B-Instruct	3B	0.665	66%
Qwen-7B-Instruct	7B	0.663	78%
LLaMA-3-8B-Instruct	8B	0.659	69%
Qwen2.5-0.5B + RAG (PowerGridQA)	0.5	**≈0.95**	**99%**

TABLE 7. NERC QA evaluation under closed-book and retrieval settings.

Model / System	RAG Used	Refusals	Numerical Fact Accuracy	Avg. Similarity
Qwen2.5-0.5B-Instruct	No	7 / 50	Failed	0.4339
Qwen-7B-Instruct	No	34 / 50	Failed	0.5004
LLaMA-3.2-3B-Instruct	No	37 / 50	Failed	0.3940
LLaMA-3-8B-Instruct	No	3 / 50	Failed	0.5150
Qwen2.5-0.5B + RAG (Reference-Locked)	**Yes**	**0 / 50**	**Correct**	**≈0.95–1.00**

The results demonstrate strong retrieval performance:

- Recall@1: The correct source document was identified as the single top hit in 82.67% of cases.
- Recall@5: The correct source document was ranked within the top 5 retrieved results in 91.15% of cases.

These high recall scores, particularly Recall@1 exceeding 80%, provide strong quantitative evidence that the structure of the NERC Operational Insight dataset effectively supports the crucial retrieval step in RAG pipelines. The inclusion of specific identifiers and relevant keywords within the questions allows standard semantic search techniques to reliably pinpoint the correct context document, validating the dataset's design for evaluating RAG systems in this specialized domain.

We emphasize that the evaluation metrics used in this work such as Exact Match (EM), F1, and BERTScore, are intended to assess language-level correctness, lexical overlap, and semantic alignment in question–answering, rather than physical system correctness or operational decision validity. EM and F1 act as strict string-based measures, while BERTScore serves as a semantic similarity proxy to capture paraphrased but conceptually relevant answers. A high

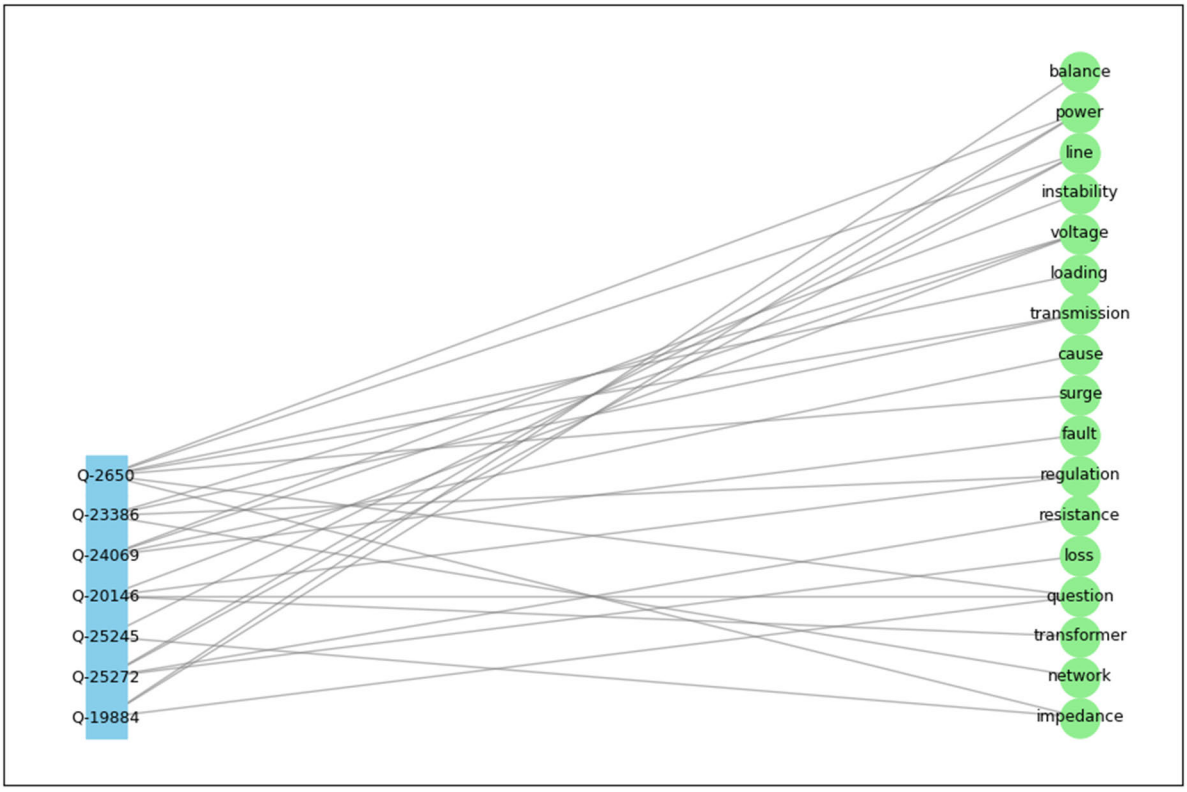

FIGURE 11. Cluster 4 represents Load and Control Systems.

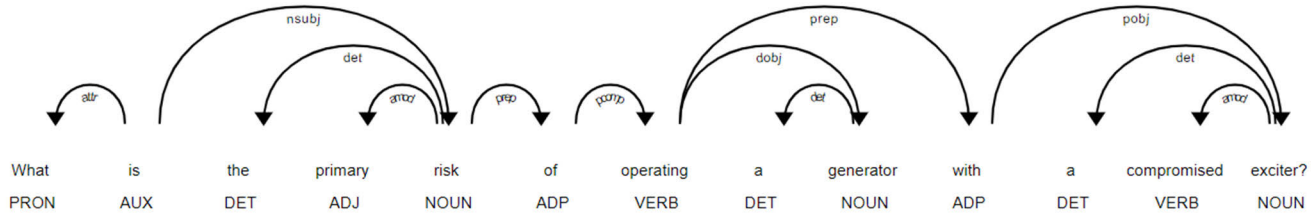

FIGURE 12. Dependency parse for question.

BERTScore therefore does not guarantee domain-correct or operationally valid responses, and should be interpreted as an informative but non-definitive signal.

V. LIMITATIONS

PowerGridQA is a static, text-based question–answer dataset grounded in authoritative power-system documentation. While it supports research on knowledge grounding, reasoning fidelity, and source-aware answering, it does not include numerical simulations, temporal system states, network topology data, or dynamic measurements. Consequently, the dataset is not intended to directly evaluate transient stability, protection performance, or real-time operational decision-making. Such evaluations require integration with power-system simulators or digital-twin environments and are left for future downstream work.

Although the current release emphasizes NERC-based operational material due to public availability, the dataset construction pipeline is source-agnostic and can be directly extended to other regional or international frameworks (e.g., ENTSO-E reports, IEC standards, or CEA documents) in future work, enabling region-specific operational subsets to be developed on top of a shared theoretical core.

In addition, PowerGridQA cannot fully capture context-dependent human operator judgment, which in real control-room environments depends on situational awareness, experience, incomplete information, and evolving system conditions. Although the dataset includes operational narratives and reasoning-based questions derived from incident reports, these scenarios are represented in a static, retrospective form rather than modeling real-time decision-making under uncertainty or human-in-the-loop reasoning. Furthermore, the dataset does not attempt to resolve

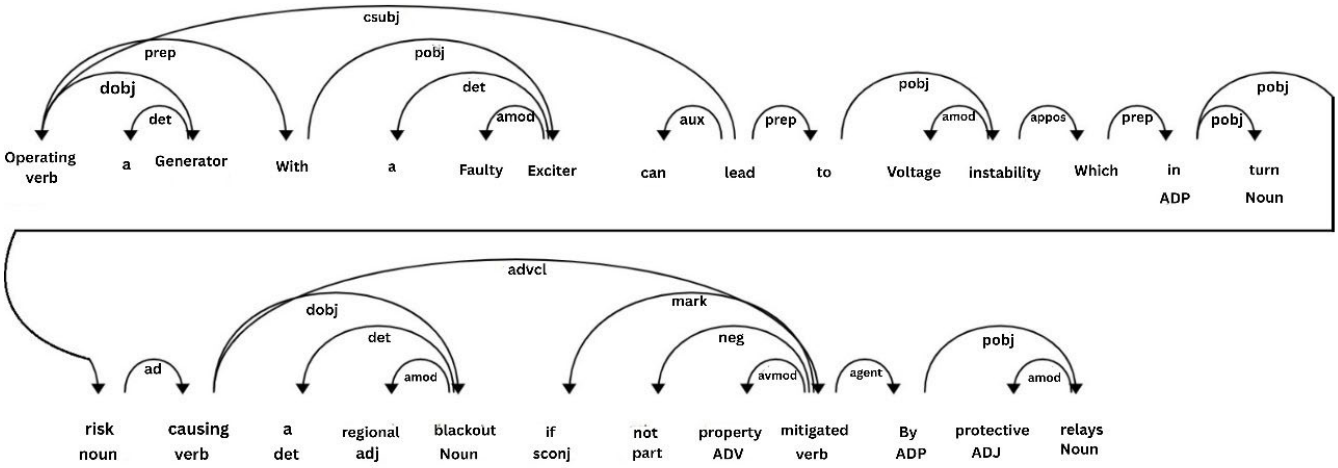

FIGURE 13. Dependency parse for Answer.

TABLE 8. Baseline performance metrics across datasets.

Metric	Power System Theory	NERC Operational Insight	Reasoning-based Case
Exact Match (EM)	0.00%	0.24%	0.00%
F1 Score	6.65%	34.54%	20.29%
BERTScore F1	83.99%	84.36%	85.40%

conflicting standards, regional practices, or alternative operational interpretations that may arise across jurisdictions or organizations; instead, it reflects the authoritative sources from which the questions are derived.

VI. DATA AND CODE AVAILABILITY

Dataset is released, and Python code used for the analysis, and validation of this dataset is publicly available on GitHub to support reproducibility and facilitate further research. The repository can be accessed at:

https://github.com/Hannaancode/PowerGridQA.git

The Dataset is available as JSONL files which can be accessed through the repository.

VII. CONCLUSION

This work shows the developing of AI systems that genuinely understand specialized domains such as power systems engineering demands benchmarks that extend beyond basic keyword matching. Addressing this gap, the dataset was designed and systematically evaluated consisting of three parts which are The Power System Theory, NERC Operational Insight, and Reasoning-Based Case, each crafted to assess distinct dimensions of comprehension: foundational theory, operational knowledge, and multi-step reasoning.

With comprehensive evaluation, encompassing lexical accuracy, semantic similarity, and structural analysis shows the distinctive strengths and purposes of each dataset. Notably, a clear difference can be seen between traditional F1 scores and higher BERTScore results, especially for the large-scale Power System Theory set. This difference underscores that true success on these datasets requires semantic understanding and contextual reasoning. The NERC Operational Insight dataset shows strong Retrieval-Augmented Generation (RAG) applications our tests have shown high retrieval accuracy and validating its effectiveness for real-world operational scenarios. Meanwhile, the Reasoning-Based Case dataset which was examined by syntactic divergence analysis, focuses on the evaluation of multi-step inference and causal reasoning.

Further thematic exploration using co-occurrence mapping and community detection visually confirmed the depth and organization of the datasets, illustrating their focused coverage of key domain concepts.

Together, this dataset has shown a rigorous and domain-grounded benchmark for evaluating AI implementation in the power systems field. Thus, encouraging the development of models capable of deeper reasoning, contextual adaptation, and reliable decision support in critical infrastructure environments.

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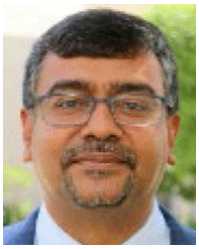

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